

Jayraj Arora

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WORK EXPERIENCE

Software Development Intern, Modernizing Medicine

Jun 2023 - Aug 2023, Aug 2024 - Present

- Automated store creation from scratch, developing backend APIs and frontend components in a complete Agile workflow.
- Processed incoming webhooks and used AWS SQS queues to update microservices, enabling real-time terminal order processing.

Graduate Teaching Assistant, California State University - Fullerton

Aug 2023 - May 2024

- Taught web backend to 80+ students in Python, MySQL and MongoDB, emphasizing API/database design and scalability.
- Guided students through practical Operating Systems concepts in Linux, utilizing hands-on examples and Linux commands.

Senior Software Developer, Oracle

Jan 2020 - Aug 2022

- Engineered Java microservices to streamline scheduling of business reports, integrating logging, security, caching and unit testing.
- Reduced rendering latency by 35% in the frontend through efficient pagination, indexed search, and dynamic filtering.
- Improved microservices deployment with CI/CD pipelines using Jenkins and Kubernetes for automated scaling and orchestration.
- Optimized 40% of Test Tracker's code using best coding practices and revamped its database design using PL/SQL and Oracle.

Software Programmer, Indiamart Intermesh

Jun 2019 - Oct 2019

- Orchestrated bulk import of micro-categories and specifications shown on the website, benefiting 150M Indiamart users.
- Achieved an 18% reduction in response time by refactoring code and optimizing database queries for a category portal.

Software Engineer, Mindfire Solutions

Nov 2017 - May 2019

- Implemented fund transfer RESTful APIs with reversal and refund mechanisms for an app used by 10K merchants.
- Migrated MySQL, Redis, and application containers to a Dockerized setup enabling seamless deployment across environments.
- Led Test Driven Development(TDD) and tested BDD, cutting bug rates by 25% and speeding up feature delivery.
- Incorporated unit testing to ensure 60% code coverage and revamped 70% of the code to improve code modularity.

Software Development Intern, iB Hubs

Jun 2017 - Aug 2017

- Built RESTful backend services in Django for an OpenAPI specification tool, supporting cross-team API design.
- Formulated and executed 175+ test cases to rigorously validate new changes in alignment with frontend requirements.

EDUCATION AND TRAINING

California State University - Fullerton

Aug 2022 - May 2024

Master of Science, Computer Science

Fullerton, California, USA

Harcourt Butler Technological University

Aug 2013 - May 2017

Bachelor of Technology, Information Technology

Kanpur, India

Publications

Shah, Nidhi, Kanika Sood, and Jayraj Arora. "Speech emotion recognition for psychotherapy: an analysis of traditional machine learning and deep learning techniques." 2023 IEEE 13th Annual Computing and Communication Workshop and Conference

Relevant Coursework

Advanced Algorithms, Web backend engineering, Advanced Database Management Systems, Data Structures, Machine Learning

PROJECTS

ThoughtBridge: Empowering Connections for Mental Health | Java, Spring Boot, Hibernate, Angular, RxJS, MongoDB

- Developed a comprehensive platform connecting therapists and patients, crafting the application layout from scratch.
- Implemented JWT authentication, caching for fast data retrieval, and queuing mechanism for asynchronous messaging.

Scalable backend architecture for Wordle game | Python, Flask, Nginx, MySQL, Litefs, Redis, RQ, Foreman, Webhooks, Linux

- Spearheaded the development of the REST API structure, game logic, database design, and three initialization scripts.
- Optimized the games microservice with load-balancing, Nginx authorization, and three DB replicas, for faster reads.
- Built a dynamic leaderboard, utilizing a Redis queue for score transmission and Redis cache for storing statistics from games.

TECHNICAL SKILLS

- **Programming Languages:** HTML, CSS, Javascript, Java, Python, Typescript, PHP, PL/SQL, Groovy
- **Frameworks:** Spring Boot, Django, Angular, React, Flask, FastAPI
- **SQL and NoSQL Databases:** MySQL, Oracle, Postgres, SQLite, Redis, MongoDB
- **Tools:** JIRA, Git, Docker, Kubernetes, Jenkins, CircleCI, SonarQube, Apache Tomcat, Nginx, Kafka, RQ, RabbitMQ, Firebase, NewRelic, Kibana, Elasticsearch, Terraform, Minikube, AWS, S3, EC2, Lambda, Cognito, Helm, Apache Spark, Hadoop, Bitbucket, Webhooks, Github Actions, IntelliJ, PyCharm, Visual Studio Code, SQL Developer, MongoDB Compass, Postman, HTTPie, Swagger
- **Other:** REST, SOAP, GraphQL, JSON, XML, SDLC, Microservices architecture, Agile Methodologies, TDD, BDD, Unit Testing